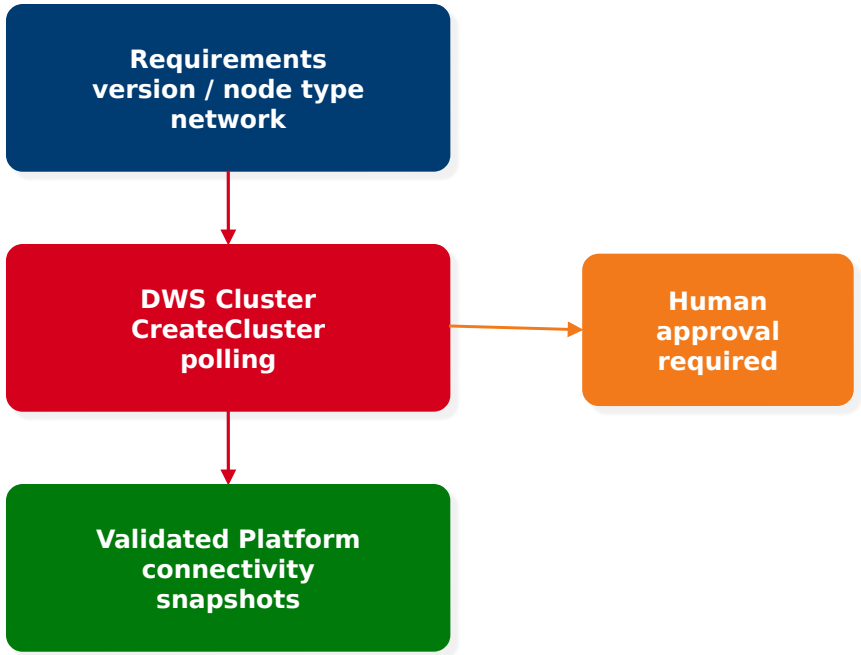
AI-Guided Huawei Cloud Scenario

# DWS Cluster Deployment

AI-guided Huawei Cloud DWS deployment using hcloud CLI, secure credentials and evidence-driven validation

READY_WITH_WARNINGS · HIGH RISK · hcloud DWS CLI



# Scenario Flow at a Glance

The scenario README orchestrates the skill, approval gates, evidence and completion criteria.

1 **Discover**

versions, node types, storage

2 **Plan**

network, capacity, credentials

3 **Deploy**

CreateCluster + polling

4 **Validate**

connectivity, security, snapshots

Primary mechanism
hcloud DWS CLI

Scenario maturity
READY_WITH_WARNINGS
Risk: HIGH RISK

Control model
Discover before create. Verify after every step. No write without explicit approval.

**Status**
**READY_WITH_WARNINGS**

**Risk**
**HIGH RISK**

**Core**
**hcloud DWS CLI**

**Control**
**approval**

# How to Use AI Safely for DWS Cluster Deployment

1

## What AI Does

Plans, analyzes, prepares runbooks, checks dependencies and explains risks. It does not replace expert review.

2

## What to Provide

Use anonymized inventories and approved metadata. Never provide real credentials, keys or customer secrets.

3

## How to Authorize

Approve each write, cutover, restore, failover or service-impacting action explicitly.

4

## When to Stop

Stop on ambiguity, unsupported region/service, unknown state, missing rollback or unclear owner.

**AI accelerates planning and evidence collection; it does not remove the need for security review, rollback planning, and human approval.**

## Your DWS Cluster Deployment Scenario

# Your Starting Point and Goal

AI must understand the business objective, not just execute isolated commands.

Your Starting Point
A Huawei Cloud DWS cluster must be designed and deployed with region-specific capabilities, secure password handling and validation.

Your Goal
Discover available capabilities, prepare network/security, create cluster with approval, validate connectivity and recovery controls.

Why AI-Assisted
AI avoids hardcoded node types, insecure credentials, over-permissive security groups and unverified cluster status.

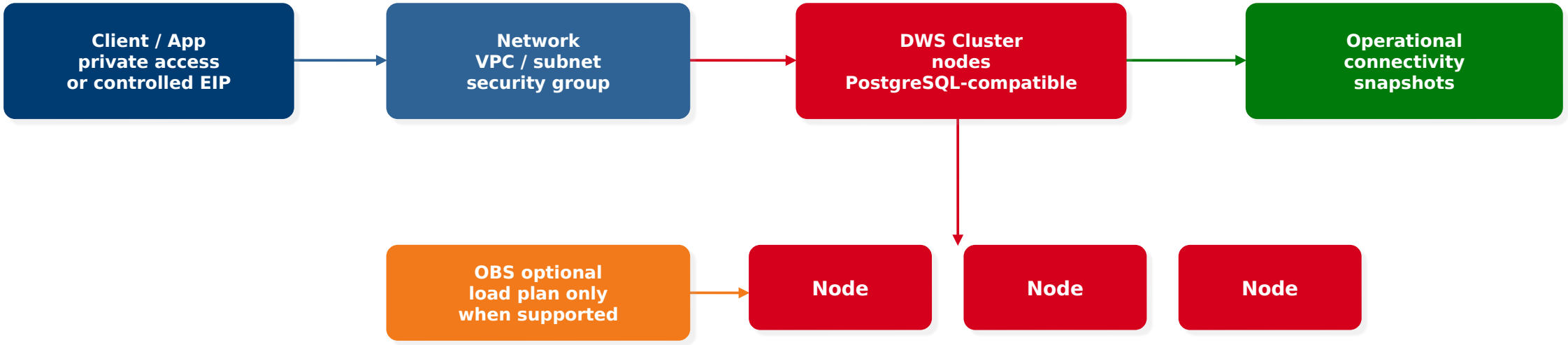
**Status**
**READY_WITH_WARNINGS**

**Risk**
**HIGH RISK**

**Mechanism**
**hcloud DWS CLI**

**Finish**
**evidence**

# Huawei Cloud DWS Deployment Architecture




**Result: proceed only after prerequisites, approvals, and verification evidence are present.**

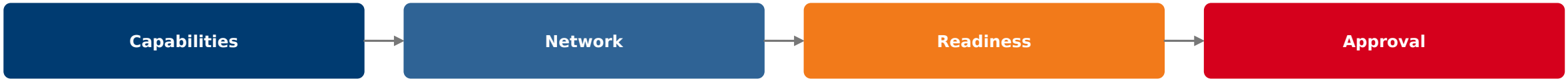
### Step 1: Ask AI to Discover and Prove Readiness

# Discover DWS Capabilities Before CreateCluster

Example prompt
"Discover DWS versions, node types, storage, AZ availability, VPC/subnet/SG, quotas, and password input options before planning."

AI should know
How to use hcloud DWS CLI read operations and avoid hardcoded versions, flavors, storage or port assumptions.

Verify
Node type, storage, subnet IPs, security group scope, quota and secure credential plan must be proven before CreateCluster.




**Stop conditions: missing target, ambiguous resource, unsupported state, missing owner, or unclear rollback.**

### Step 2: Ask AI to Prepare the Controlled Plan

# Prepare Cluster Plan and Secure Inputs

**Inputs**

**Plan**

**Approval**

## Readiness evidence the agent must produce

- ✓ Version/node/storage discovered
- ✓ Subnet has sufficient capacity
- ✓ No 0.0.0.0/0 inbound
- ✓ Credential handling approved

## Plan elements

### Capacity plan

Node type, node count, storage, HA assumptions, growth and budget.

### Network plan

VPC, subnet, security group, optional EIP, allowed CIDR only.

### Credential plan

No password in shell history; use secure JSON/input mechanism.

**Every plan must identify approval owner, verification evidence, and rollback or recovery behavior.**

### Step 3: Ask AI to Execute One Controlled Phase at a Time

# Execution: Controlled Two-Phase Process

## Phase 1: Create and Poll

CreateCluster only after approval
Capture request/cluster ID
Poll Show/List status
Handle timeout or failed state

**Ready** →

## Phase 2: Configure and Validate

Optional EIP with authorized CIDR
Connect psql/JDBC
Prepare DB/schema plan
Configure snapshots




*On failure: preserve IDs/logs/evidence → classify root cause → resume only from verified checkpoint.*

### Step 4: Ask AI to Handle Errors and Edge Cases

# DWS Deployment Error Handling

| | | |
|---|---|---|
| N | **Node type unavailable** | Re-discover capabilities; do not hardcode alternate flavor. |
| S | **Subnet insufficient** | Stop and resize/select subnet before creation. |
| P | **Password exposure risk** | Regenerate securely; never store in repo or shell history. |
| C | **Cluster failed/timeout** | Preserve request ID and status; do not auto-delete. |
| E | **EIP requested** | Only allowed CIDR; never 0.0.0.0/0. |

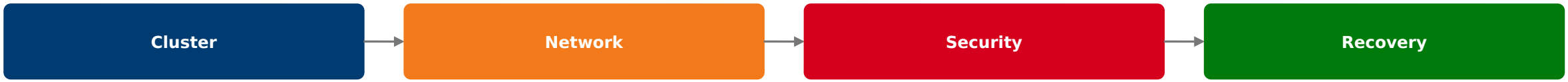
**Verify and prevent hallucinations: when AI says “fixed”, require concrete evidence — logs, status, checks, and owner validation.**

### Step 5: Ask AI to Validate Before Closure

# Operational Readiness Must Be Evidence-Based

Do not accept generic “looks good”. Require proof before closure.

- ✓ Cluster status and configuration match plan
- ✓ Endpoint and port verified from cluster data
- ✓ Security group and optional EIP approved
- ✓ Connectivity test planned/executed safely
- ✓ Snapshot and recovery plan recorded



## Equip Your AI Assistant

# What the AI Needs to Execute the Scenario

Start from scenario README; load only the required skill for the current phase.

### Scenario README

Scenario README scopes the deployment, completion criteria and approval gates.

### Operational Skill

Skill uses hcloud DWS CLI for lifecycle with secure input and polling.

### Shared Components

Deploy MCP may help prerequisites only; DWS lifecycle remains CLI. Capability builder tracks future extension.

**Rule of thumb: do not load every skill; identify the current phase, load the relevant SKILL.md, execute one controlled step, verify, then return to the scenario README.**

## Pilot to Production Journey

# Recommended DWS Deployment Journey

**This scenario is complete only when its objective-specific validation criteria are met and closure is approved.**

